

Alternative transport architectures for core ATM networks

P A Veitch, P R Richards, P J McCartney and D Johnson

In order to remain competitive in an increasingly aggressive market-place, it is essential that network operators deliver cost-effective, resilient bulk transport. This will enable the unit costs of circuit delivery in the core network to be reduced, while assuring high-level quality of service. In particular, the growth in advanced data services which exploit ATM technology will place significant demands on the core transmission layer of telecommunications networks. This paper investigates alternative transport architectures for core ATM networks.

1. Introduction

Due to huge competitive pressures from fellow telcos, cable companies and Internet service providers (ISPs), companies like BT have to ensure that traditional services like voice and private circuits (PCs), as well as emerging data services, are supported by a cost-effective, resilient bulk transport layer. This will enable the unit costs of circuit delivery in the core network to be reduced, while assuring high-level quality of service. Of particular concern is the imminent arrival of the 'data wave', a popular term which describes the anticipated explosion in the usage of data-oriented services. The Internet is estimated to be growing at 10–15% per month [1], which is not surprising considering the everyday use of applications which exploit TCP/IP such as e-mail (a growing proportion of messages contain bandwidth-consuming attachments), and World Wide Web (WWW) browsing/downloading.

In addition to TCP/IP-based applications, other data services such as Frame Relay (FR) and Switched Multi-megabit Data Service (SMDS), as well as 'native' ATM services like BT's CellStream service, may exploit a common multiservice platform (MSP) based on ATM technology. Although ATM can in theory be used to support all sorts of applications including voice and video, its principal use in the near future will be to support data services. Consequently, the growth in advanced data services which exploit ATM technology will place significant demands on the core transmission layer of telecommunications networks.

This paper investigates alternative architectures for ATM transport in the inner core which is the high-speed backbone network that interconnects large switches in the main cities. The key drivers for connecting ATM switching equipment over a synchronous digital hierarchy (SDH)

transport layer are explained in section 2. Subsequently, two distinct SDH structures will be described, namely a mesh of digital crossconnect systems (DXCs), and a shared protection ring (SPRing) architecture. A cost analysis demonstrates the economic benefits of the SPRing architecture over the mesh. Section 3 presents a comparison of the SDH-based SPRing architecture for ATM switch interconnection, with a 'direct fibre (DF)' architecture, whereby ATM switches are connected directly with optical fibre. A cost comparison demonstrates the distance-dependence of the result, with the direct fibre structure being shown to be more amenable to shorter 'metro' distances. In addition to cost, the implications of a direct fibre architecture on network resilience are examined. Conclusions are drawn in section 4.

2. SDH transport for ATM

2.1 Why ATM-over-SDH?

Most commercially operated ATM networks are underpinned by a synchronous digital hierarchy (SDH) transport network. SDH (or SONET in the US¹) networks have been rolled out by many network operators in Europe, Japan and North America since the early 1990s. SDH is a sophisticated multiplexing and transmission protocol which provides many benefits, especially when compared with the plesiochronous digital hierarchy (PDH) mode of transport:

- lower order channels at 2 Mbit/s and above can be dropped and inserted directly into the fundamental SDH line rate of 155 Mbit/s by pointer processing avoiding the need for costly and space-consuming 'multiplex mountains',

- multiplexing of the line rates to higher rates (for example, 155 Mbit/s to 622 Mbit/s) is performed by simple byte interleaving,
- substantial overheads are embedded in the signal hierarchy enabling enhanced manageability of equipment — this gives SDH an edge in terms of automatic circuit configuration and advanced resilience mechanisms,
- as well as being backwards compatible with PDH circuit bit rates, SDH can support broadband circuit bit rates at 155 and 622 Mbit/s using concatenation of tributary rates.

Hence, since many operators have well-established, resilient SDH transport networks which are capable of directly supporting broadband circuit interfaces at rates of 155 Mbit/s and 622 Mbit/s, it has been deemed sensible from an economic and management perspective to deploy ATM-over-SDH.

2.2 Alternative SDH transport architectures

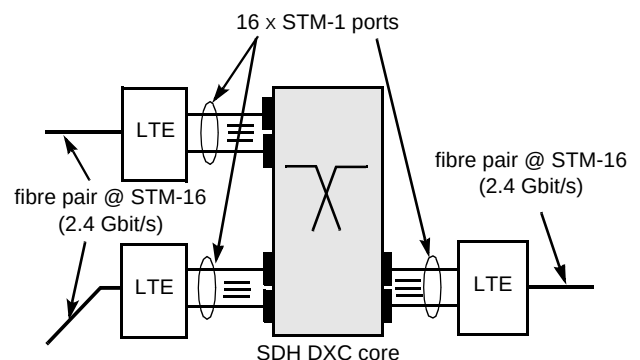
When most network operators commenced deployment of SDH transport networks in the early 1990s, digital crossconnect systems capable of switching tens of Gbits of traffic were the most suitable building blocks with which to construct a large-capacity meshed backbone network [2—4]. Recent advances in high-speed electronic add/drop multiplexing (ADM) equipment with shared protection functionality has stimulated interest in the prospect of ring deployment in backbone networks. ADM rings have already been deployed extensively in local and metro regions of networks with line speeds of up to STM-4 rate (622 Mbit/s) [2—4]. The availability of ADM equipment at STM-16 (2.4 Gbit/s) and STM-64 (10 Gbit/s) rates, as well as the emergence of wavelength division multiplexing (WDM) products enabling 40 Gbit/s transmission capacity in a single fibre [5], suggests there may be a more economic alternative to meshed backbones based on DXCs.

The basic architecture of an SDH DXC node includes the DXC switching core, STM-N ports/interfaces on the DXC, and line termination equipment (LTE). Internal switching/crossconnecting may be performed at the 2 Mbit/s virtual container (VC-12) level, with such devices typically being deployed where a lot of 2 Mbit/s circuits must be switched. Alternatively, switching at the 140 Mbit/s virtual container (VC-4) level may be performed at bulk routing nodes to support transit switching of large demands, while acting as flexibility points to assist with restoration and planned maintenance activity. Bulk routing nodes containing VC-4 DXCs will often be accompanied by STM-16 LTEs enabling transmission rates of 2.4 Gbit/s per fibre pair as shown in Fig 1(a). Fibre pairs can be terminated (via LTEs) on to as many ports as the switch can handle; if a DXC has a 40 Gbit/s switching capacity, thus accommodat-

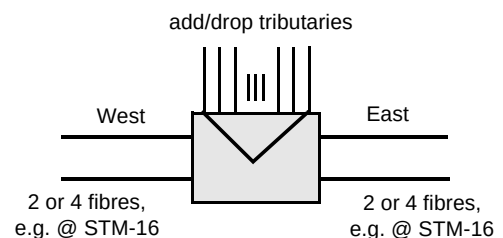
ing 256 STM-1 ports, up to 16 fibre pairs at 2.4 Gbit/s (STM-16) transmission rates can be terminated at a single DXC site.

The basic ADM architecture is shown in Fig 1(b). Tributaries enter the ADM core where switching will either be at the tributary rate or at a finer level of granularity. For example, STM-1 tributaries may be supported in an STM-16 ADM, with internal switching at either VC-12 or VC-4 granularity. This will depend on the mix of lower-level granularity traffic sourcing and sinking at an ADM site, and the bulk VC-4 transit traffic at that site. As the ADM is configured in a ring, traffic is routed either in the east or west directions, depending on which node of the ring the traffic has to reach. There are functional differences between 2-fibre and 4-fibre ADM rings related to traffic routing and protection switching complexity which are outside the scope of this paper; however, Wu [6] provides a detailed insight into the various architectures. The ADM and DXC architectures differ in the following ways:

- line termination functionality is integrated into the ADM equipment, while discrete LTEs are required with DXCs,
- there are several ways to access a fibre route from a DXC, and only two from an ADM.



(a) SDH digital crossconnect (DXC) structure with STM-16 line system



(b) SDH add drop multiplexer (ADM) structure with STM-16 interfaces

Fig 1 Comparison of typical DXC and ADM nodal architectures.

2.3 Resilience of SDH mesh and ring architectures

Resilience is an essential feature of backbone architectures carrying huge volumes of data, to counteract many threats to the network's integrity such as cable breaks and node failures [7]. Different resilience mechanisms can be supported in mesh and ring architectures.

Mesh networks based on DXCs can support 1+1 sub-network connection protection (SNCP) of circuits with receive-end protection switching being performed by ADMs in around 50 ms, or switching may be carried out by DXCs themselves [8]. In any case, this is a very costly method of protection as spare resources are totally dedicated to specific circuits, and cannot be shared. Mesh restoration with shared spare capacity is an economic alternative to 1+1 protection, with only 40–50% of the working capacity being necessary to provide resilience from any single cable break in a well-connected mesh (e.g. connectivity factor between 3 and 4) [8]. The disadvantage of this technique is that restoration is typically realised by a centralised management system which has a global view of failed working circuits and whatever spare capacity exists to reroute such circuits [9,10]; this can take several minutes, compared with sub-second recovery achieved with protection switching.

ADMs may be configured as dedicated protection rings (DPRings) where protection capacity is reserved in a 1+1 fashion, or they may be configured as shared protection rings (SPRings) where sufficient spare capacity is provisioned and shared to enable recovery from any single cable break. Although SPRings make more efficient use of resources than DPRings, they are more costly due to the more complex software required to co-ordinate protection switching. Whatever ADM technology is exploited, protection switching in self-healing ring architectures can be accomplished in 50 ms [6].

2.4 Cost comparison of alternative SDH transport architectures

If a network operator wishes to build an SDH backbone network for bulk transmission (whether it be from scratch for a 'greenfield' design, or by re-design of an existing legacy network), the comparative costs of alternative architectures must be well understood. This section presents an analysis of the cost trade-offs between mesh and ring architectures for core SDH transmission. Analyses have shown that communities-of-interest exist between nodes within a regional sub-network which is part of a larger, backbone network. In other words, a significant proportion of traffic arriving at a backbone node will be routed towards neighbouring nodes as shown in Fig 2.

The characteristic of core network traffic depicted in Fig 2 can be exploited by deploying rings to capture communities of interest that exist within geographical sub-

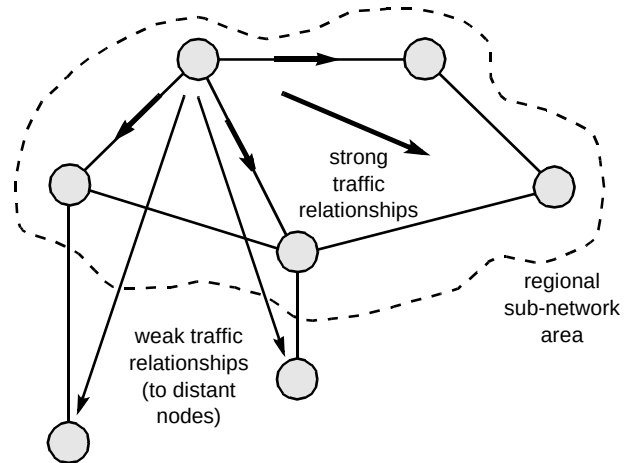


Fig 2 Dependence of traffic on geography.

network regions. To this end, a sub-network of six nodes as shown in Fig 3 was used as the basis of a cost analysis to compare mesh and ring architectures.

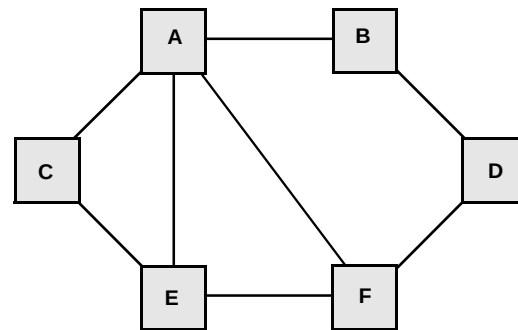


Fig 3 Six-node sub-network employed in cost analysis.

For the cost modelling, it was assumed that:

- the mesh will be configured in the manner shown with eight inter-nodal links in total, namely, A-B, A-C, A-E, A-F, B-D, C-E, D-F and E-F,
- the ring design will comprise a single ADM ring encompassing the six nodes shown with inter-nodal links, namely A-B, A-C, B-D, C-E, D-F and E-F,
- the sub-network is part of a 40-node backbone network, approximating to the scale of BT's inner core SDH network,
- a linear increase in traffic reaches the inner core network over a period of 5 years, commencing in 1998.

The cost analysis considers mesh and ring architectures comprising a variety of technologies. In all cases, the aim was to dimension a network design capable of supporting a VC-4 demand matrix, thus representing a bulk transport network.² For the mesh, 40 Gbit/s 4/4 DXCs with STM-1 ports were assumed to operate in conjunction with either

² This implies that the costs of service switches (e.g. 4/1 DXCs, ATM, PSTN) are omitted.

STM-16 LTEs or STM-64 LTEs. 1+1 diverse routing of VC-4 demands was assumed in the mesh, with protection switching realised at the DXC (i.e. there are no additional ADMs). Two types of self-healing ADM ring architectures were evaluated:

- STM-16 2-fibre shared protection rings,
- STM-64 2-fibre shared protection rings.

Where necessary, ADM rings are stacked to accommodate all the traffic in the sub-network.

Figure 4 shows the average costings for each architecture from 10 samples of randomly varied traffic within the sub-network region. Core traffic predictions have been based on current narrowband services such as switched voice, as well as future broadband data and video services which will originate from a client ATM layer. By year 5, the STM-16 SPRing architecture is 38% less costly than the protected mesh with STM-16 line systems while the STM-64 SPRing is 30% less expensive than the protected mesh with STM-64 line systems. Hence, it is clear that cost benefits accrue by deploying ring architectures rather than continuing with mesh deployment for growth traffic, a result that holds for both uniform and non-uniform traffic spread within a community-of-interest sub-network. Savings are made largely due to the architectural characteristics and cost of ADM rings compared with mesh architectures. For example, much of the mesh cost resides in LTEs and associated tributaries, a cost not incurred in any of the SDH ring transport solutions. Other studies have confirmed the economic advantages of SPRings over mesh architectures for larger network configurations and a variety of traffic distributions [11].

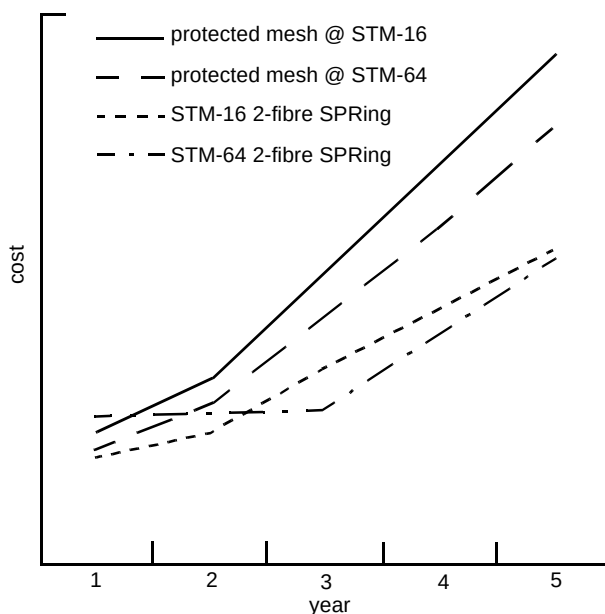


Fig 4 Cost comparison for non-uniform traffic over 5-year period.

3. Direct fibre versus SPRings for ATM switch interconnect

3.1 Basic architecture

Although it makes operational sense to interconnect ATM switches via a readily accessible SDH transport platform (Fig 5(a)) with in-built resilience (whether it be with 1+1 protection in a mesh architecture or with shared protection switching achieved in a SPRing architecture), economic benefits may accrue by exploiting direct fibre (DF) to interconnect ATM switches. This architecture is illustrated in Fig 5(b) which shows fibre connections installed directly between STM-4 (622 Mbit/s) optical ports on the ATM switches. It is worth pointing out that since STM-N interfaces exist on the ATM switches, point-to-point SDH transport of ATM cells is employed. However, the lack of any switched networking at the SDH layer means SDH resilience mechanisms cannot be employed. It is thus assumed that 1+1 port protection is performed by the interfaces on the ATM switches.

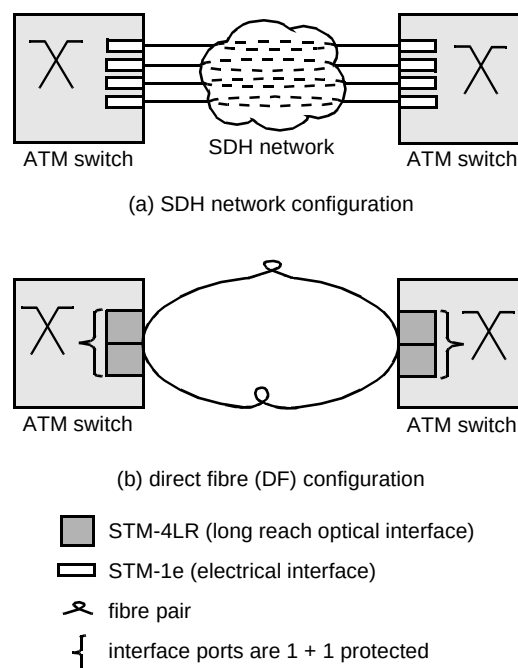


Fig 5 Alternative architectures for links between ATM switches.

3.2 Cost analysis

To investigate the economics of the direct fibre method of switch interconnect with the SDH-based SPRing architecture, the same six-node network model as before was employed. As shown in Fig 6, the distance between nodes was allowed to vary in order to test the sensitivity of the cost comparison on this important parameter. ATM switches were assumed to exist at each nodal location. The underlying transport can either be SPRing-based, whereby a

single ADM ring encompasses the six nodes shown with inter-nodal links A-B, A-C, B-D, C-E, D-F and E-F, or it can be DF-based whereby the underlying mesh structure is used to provide working and protection routes between nodes. There are two distinct variants on the DF architecture, depending on whether traffic is switched at intermediate ATM nodes, or 'glassed through'. With transit switching, a fibre carrying a traffic demand between nodes A and D will be terminated at node B such that ATM cells are switched and then transmitted on a separate fibre between nodes B and D. The 'glass-through' architecture will involve a single fibre pair connection which routes demands between A and D, following a common physical duct infrastructure which passes through node location B. These variants are designated *DF(tr)* and *DF(gt)* respectively.

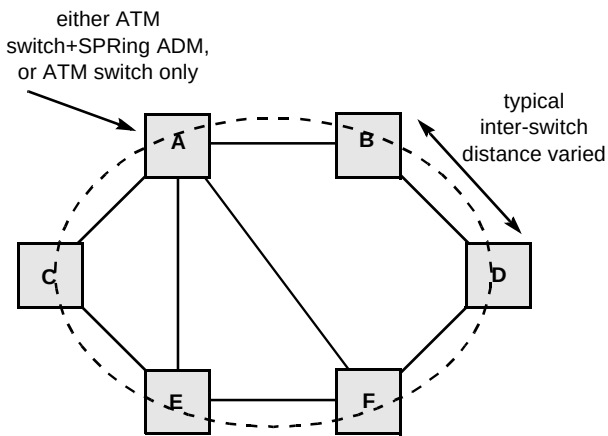


Fig 6 Network model for DF versus SPRing cost comparison.

Figure 7 shows the average capital costs of each architecture modelled with a distribution of predicted growth traffic carried over the ATM multi-service platform (MSP), versus the assumed distance to exist between each node.

The graph indicates that direct fibre interconnect with transit switching is always the most costly option. Meanwhile, a break-point exists between direct fibre without transit switching and the SPRing architecture. At very short inter-switch distances, the transport costs are dominated by switching equipment costs. Since the direct fibre structure includes just ATM switches, whereas the SPRing architecture involves ATM switches as well as SDH ADMs, the latter proves more costly. With increasing distance between the nodes, the dominant cost pertains to transmission equipment such as fibre and regenerators. The direct fibre architecture becomes increasingly more expensive than the SPRing structure due to the fact that transmission is at STM-4 speeds compared with the STM-16 line speeds of the SPRing ADMs. If a traffic demand of 4*STM-4s was required between two nodes therefore, four

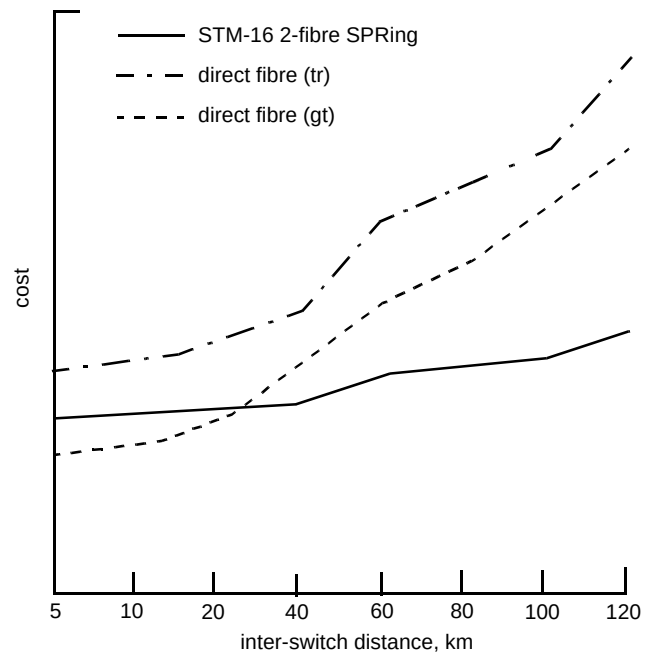


Fig 7 Cost analysis of direct fibre (DF) versus SPRings for ATM switch interconnect.

fibre pairs would be needed in the DF model compared with two fibre pairs for the SPRings³.

It is important to stress the sensitivity of the break-point between the SPRing and direct fibre (gt) architectures, shown in Fig 7, to the surrounding assumptions. For example, lower SDH ADM costs would favour the SPRing architecture for even shorter inter-switch distances than those shown. On the other hand, reduced transmission costs yielded by higher capacity transmission systems based on wavelength division multiplexing (WDM) techniques, or indeed with higher speed interfaces on the ATM switches such as STM-16, would shift the break-point in favour of the direct fibre architecture for greater inter-switch distances. All of these factors must be fed into further studies to determine the suitability of direct fibre interconnection for ATM transport. In the meantime, it is probably fair to say that direct fibre interconnect is not suitable for inner core architectures; however, it could be considered in relatively small 'metropolitan' environments.

3.3 Resilience implications of direct fibre inter-connection

Since direct fibre interconnection architectures bypass the networked SDH layer, the ATM layer is consequently responsible for delivering network resilience to cable breaks and node failures. Although 1+1 port protection

³ As half the capacity in each SPRing section is reserved for protection, there is effectively 8*STM-1 worth of usable capacity.

mechanisms may be supported on ATM switches, this may not be the most desirable form of resilience for a multi-service platform supporting a wide range of services. Two reasons for this are:

- cost — although 1+1 protection on ATM switches can be executed in the absence of a switched SDH network layer, the fact remains that 1+1 protection at any transport layer is resource-hungry and thus costly,
- resilience requirements — not all services supported on an ATM-based MSP will require fast protection switching, but may make do with slower, and less costly, resilience mechanisms executed in the ATM layer.

The relationship between resilience requirements of services supported by ATM, and protection and restoration mechanisms, which may be applied to satisfy those requirements, are discussed in detail elsewhere [12].

4. Conclusions

In order to remain competitive in a market-place which is experiencing huge growth in data services alongside traditional services like voice, it is essential that network operators deliver cost-effective, resilient bulk transport. This paper has investigated alternative inner core transport architectures to satisfy the growing demand in services which exploit an ATM-based multiservice platform. Regarding SDH as an underpinning transport layer, it has been shown that:

- continuing with mesh architectures based on DXCs and 1+1 protection may incur unnecessarily high transport costs compared with alternative, ring-based architectures,
- capital cost savings of up to 38% are possible for an STM-16 shared protection ring (SPRing) architecture compared with a 1+1 DXC-based mesh.

Subsequently, the possibility of exploiting direct fibre (DF) interconnection of ATM switches, whereby no SDH networking equipment is employed, was examined, from which the following key conclusions can be drawn:

- economic benefits of direct fibre interconnection are only accrued at very short inter-switch distances, making such architectures more suitable for metropolitan, rather than inner-core environments,

- bypassing a resilient SDH layer shifts the onus of resilience to the ATM layer — although initially 1+1 port protection methods may be used, more resource-efficient shared-protection techniques should, however, be feasible in the future.

Further work must address how best to combine IP, ATM, SDH and WDM technologies to deliver the most cost-effective transport of current and future services.

References

- 1 Ovum report: 'The future of broadband networking: TCP/IP versus ATM', <http://www.intellact.nat.bt.com/intellact/reports/ovum/tcp/index.htm> (January 1998).
- 2 Passeri P et al: 'Introducing SDH systems in Europe', IEEE, LTS, pp 38—43 (November 1991).
- 3 Shirakawa H, Maki K and Miura H: 'Japan's network evolution relies on SDH-based systems', IEEE, LTS, pp 14—18 (November 1991).
- 4 Ching Y and Sabit H S: 'SONET implementation', IEEE, Communications Magazine, pp 34—40 (September 1993).
- 5 Kidd P, Dhoooper J and Allen L: 'Widening the fibre highway', Telecommunications, pp 50—56 (November 1997).
- 6 Wu T H: 'Fiber network service survivability', Artech House (1992).
- 7 McDonald J C: 'Public network integrity — avoiding a crisis in trust', IEEE, J-SAC, pp 5—12 (January 1994).
- 8 Johnson D: 'Survivability strategies for broadband networks', IEEE, Globecom '96, London, pp 452—456 (November 1996).
- 9 Yamagishi K, Sasaki N and Morino K: 'An implementation of a TMN-based SDH management system in Japan', IEEE, Communications Magazine, pp 80—95 (March 1995).
- 10 Chao C W, Fuoco G and Kropfl D: 'FASTAR platform gives the network a competitive edge', AT&T Technical Journal, pp 69—81 (July/August 1994).
- 11 Johnson D, Veitch P A and Hayman N: 'Core transport network redesign', Submitted to 1st International Workshop on the Design of Reliable Communication Networks, Bruges (1998).
- 12 Veitch P A and Johnson D: 'ATM network resilience', IEEE, Network, pp 26—33 (September/October 1997).



Paul Veitch joined BT in 1996 after graduating from the University of Strathclyde with MEng and PhD degrees in Electronic and Electrical Engineering. He has been involved in the design and analysis of resilient transport network infrastructures encompassing technologies such as ATM, SDH and WDM.

He currently works in the Transport Design and Performance unit, and is assessing the economic and operational implications of alternative outer core transport architectures.



Peter Richards joined BT as an apprentice in 1977 and graduated from Essex University with a BSc degree in Electronic Engineering. He joined BT Laboratories in 1986 to design and test integrated circuits and crossconnect systems for the early studies of SDH transmission. He moved on to the design and analysis of SDH networks and, later, broadband networks.

He has represented BT on SDH issues at the ITU and ETSI and worked on European collaborative projects. Recently he has been involved with broadband network architecture design, including switch and PON head-end node selection and core transport design for the ATM infrastructure project.

His current work involves investigating future transport network architectures for the multi-service broadband platform and the evaluation of IP networks and routers.



Pam McCartney joined BT Laboratories as an apprentice in 1977, and graduated from Southampton University with a BSc degree in Electronic Engineering in 1984. Since then she has worked in the field of digital transmission systems, which has included the design and development of a 2 Mbit/s to 140 Mbit/s synchronous multiplexing equipment (BTSMUX 1401 product), SDH network planning and design, ISDN, and Kilostream ACE network studies. More recently, she has been involved with broadband network architecture design, including switch and PON head-end node selection and core transport design for ATM networks. She is currently leading a task investigating future transport network architectures for the multi-service broadband platform.



Dave Johnson joined BT in 1986 after graduating from Lancaster University with a first class honours degree in Physical Electronic Engineering. He worked for five years on systems reliability, including the reliability analysis of network architectures and the production of computer software for system reliability prediction. He then spent six months working on intelligent networks. In 1991 he moved on to the research of novel resilience mechanisms for telecommunications networks and the development of network resilience planning tools, resilience strategy, and a resilience evolution plan for BT. He currently leads a small team in the Network Transport Engineering Centre, analysing and planning changes to the structure of BT's transport network.

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